



TEST CODE **02207020**

FORM TP 2013147

MAY/JUNE 2013

C A R I B B E A N E X A M I N A T I O N S C O U N C I L

C A R I B B E A N A D V A N C E D P R O F I C I E N C Y E X A M I N A T I O N[®]

BIOLOGY

UNIT 2 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions in two sections. Answer ALL questions.
2. For Section A, write your answers in the spaces provided in this booklet.
3. For Section B, write your answers in the spaces provided at the end of each question in this booklet.
4. The use of silent non-programmable calculators is allowed.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

SECTION A

Answer ALL questions.

Write your answers in the spaces provided in this booklet.

1. (a) Figure 1 is a flow chart which represents a part of the glycolysis pathway.

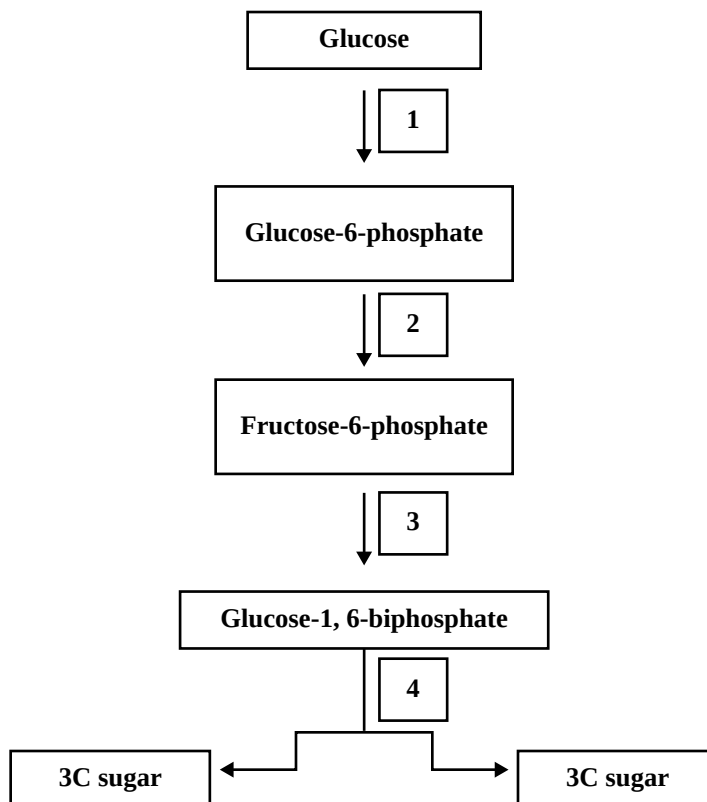


Figure 1. Part of the glycolysis pathway

- (i) In which part of the cell does the process of glycolysis occur?

[1 mark]

- (ii) State what is occurring at EACH of the stages labelled 1, 2, 3 and 4 in Figure 1.

1: _____

2: _____

3: _____

4: _____

[2 marks]

- (iii) Suggest the significance of the pathway shown in Figure 1.

[2 marks]

- (b) Figure 2 represents a general model for nutrient cycling in an ecosystem highlighting the main reservoirs of elements. Arrows indicate the processes that move nutrients between reservoirs.

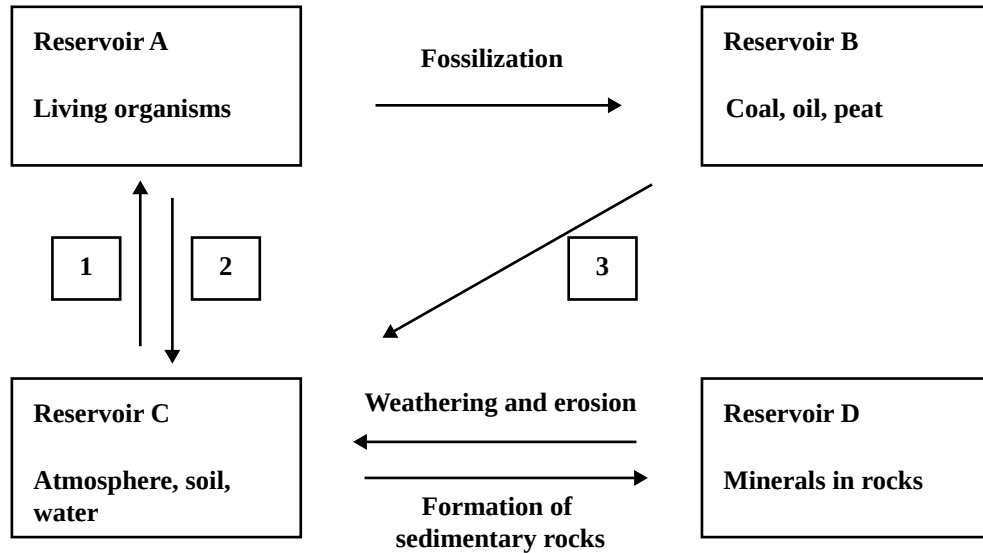


Figure 2. A general model for nutrient cycling

(Source: Adapted from Biology 2009 Campbell and Reece 8th Ed.)

- (i) Deduce which reservoir has
- organic** materials which are **available** as nutrients
 - inorganic** materials which are **unavailable** as nutrients.
- [2 marks]
- (ii) Name the processes highlighted by the arrows labelled 1, 2 and 3 in Figure 2, which must occur to transfer nutrients between reservoirs.
- 1: _____
- 2: _____
- 3: _____
- [3 marks]
- (iii) Recent evidence suggests that mycorrhizal fungi can release acids that dissolve some minerals. Use an arrow and the letter M to indicate on Figure 2 where this fungal activity fits into the model.
- [1 mark]

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- (c) Figure 3 represents the effect of light intensity on rate of photosynthesis.

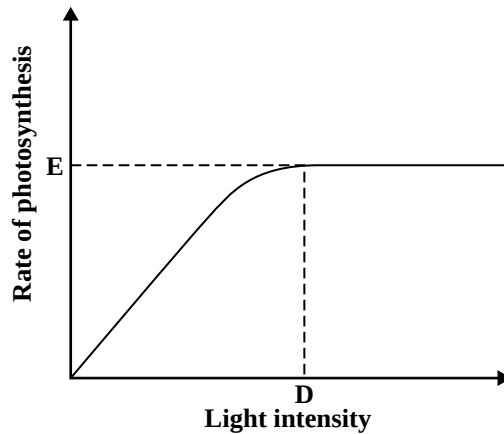


Figure 3. Effect of light intensity on rate of photosynthesis

- (i) Use labelled double-headed arrows ($\leftrightarrow$) on the graph in Figure 3 to identify the region where
- light is a limiting factor
 - other factors become limiting.
- (ii) What is the significance of EACH of the points, **D** and **E**, on the axes of the graph in Figure 3?

Point D:

Point E:

[2 marks]

Total 15 marks

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2. (a) Figure 4 is a photomicrograph of a stained, longitudinal section of vascular tissue showing phloem and xylem vessels.

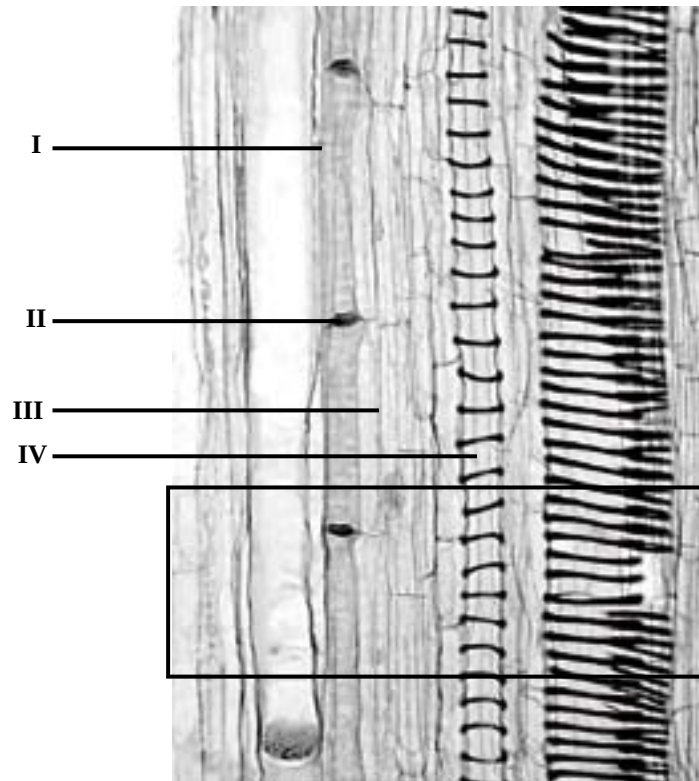


Figure 4. Photomicrograph of longitudinal section of vascular tissue

(Source: <http://www.wellcome.ac.uk/en/bia/gallery.htm>)

- (i) In the space below, make a detailed drawing of the region highlighted by the box in Figure 4. Your drawing should be the actual size shown in Figure 4.



[5 marks]

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- (ii) Identify the structures highlighted by the lines labelled I, II, III and IV in Figure 4.

I _____

II _____

III _____

IV _____

[2 marks]

- (b) Figure 5 shows the relationship between gender and age in an Arab population with a high cardiovascular risk profile.

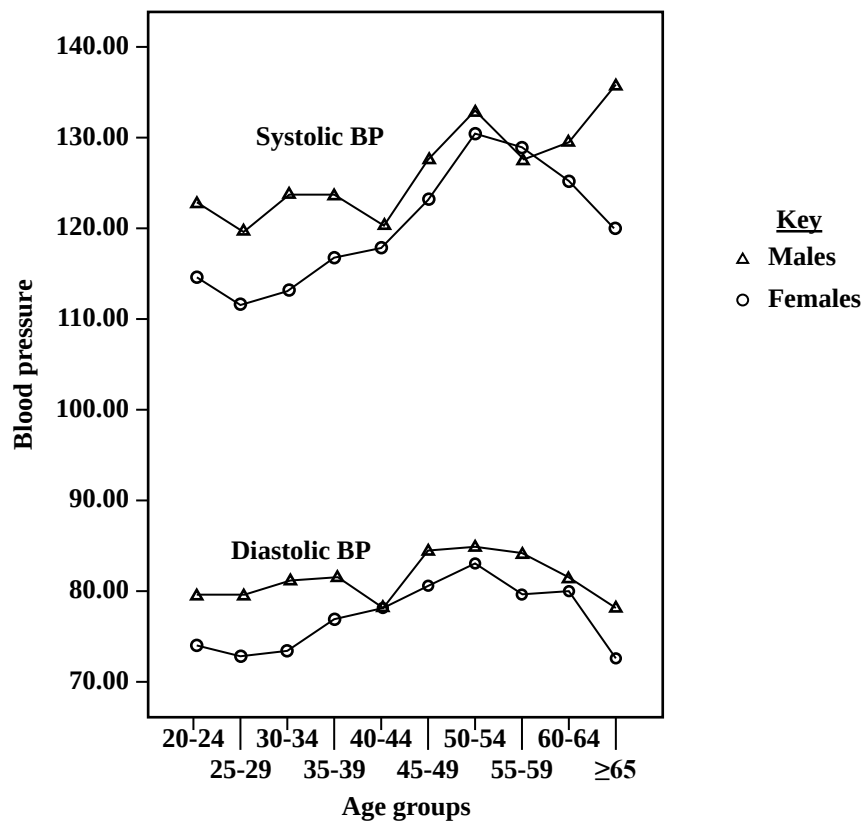


Figure 5. Gender influence on blood pressure and change in blood pressure with age

(Source: <http://www.cardiab.com/content/8/1/21/figure/F2?highres=y>)

- (i) Distinguish between the terms 'systolic' and 'diastolic' blood pressure.

[2 marks]

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- (ii) Comment on the **overall** trend in changes in systolic and diastolic blood pressure of the population with increasing age.

Systolic: _____

Diastolic: _____

[4 marks]

- (iii) What deduction can be made about the relationship between gender and blood pressure?

[2 marks]

Total 15 marks

3. (a) Table 1 shows mortality rates for some of the most common cancers in men and women of Latin America and the Caribbean in the year 2008.

TABLE 1: MORTALITY RATES FOR MEN AND WOMEN IN LATIN AMERICA AND THE CARIBBEAN. (RATE IS PER 100 000.)

Cancer	Men	Women
Breast	0	12.4
Cervix/uterus	0	10.8
Colorectum	7.1	6.1
Liver	5.9	4.8
Lung	17.7	7.3
Prostate	16.3	0
Stomach	12.8	6.8

(Source: www.paho.org/english/ad/dpc/nc/pcc-fact-sheet-LAC.pdf)

- (i) On the grid provided on page 11, construct a histogram of the mortality rates for both men and women using the data given in Table 1. **[6 marks]**
- (ii) Based on the data presented in Table 1, comment on which type of cancer is the leading cause of death for men and women, and for both sexes combined. Quote the mortality rate in your answer.

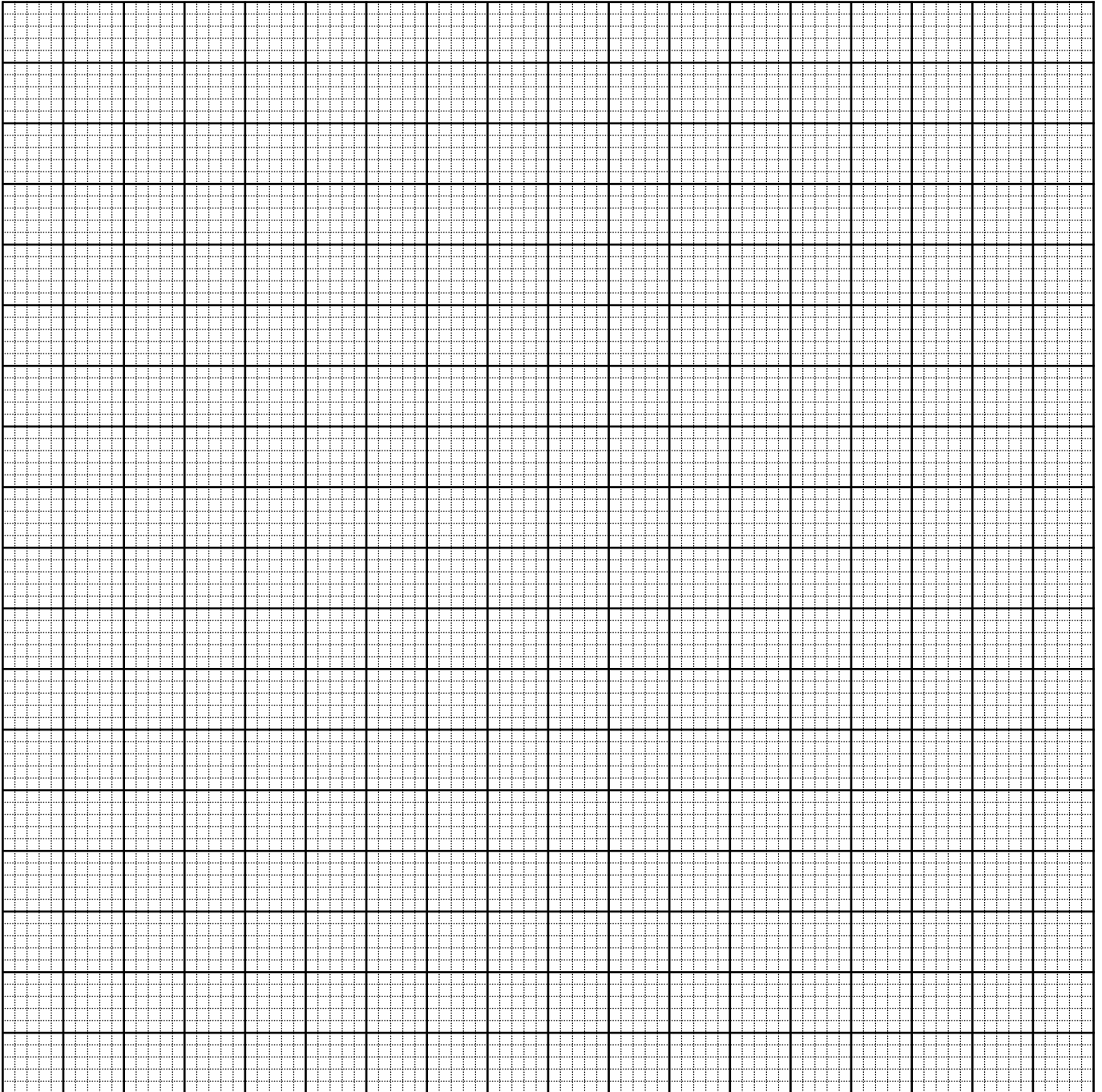
Men: _____

Women: _____

Both sexes: _____

[3 marks]

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- (b) Figure 6 shows the general relationship between exercise intensity and oxygen consumption.

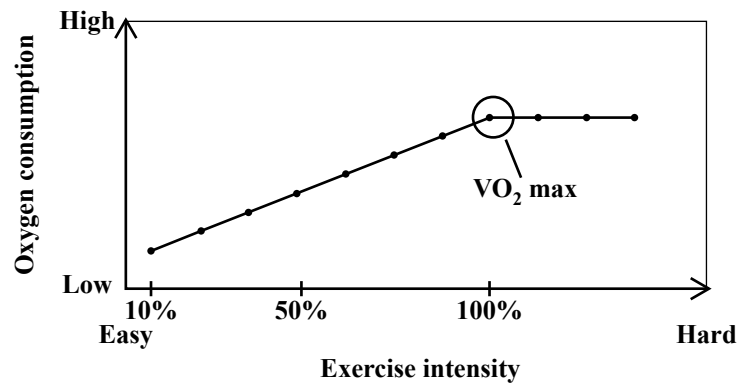


Figure 6. Oxygen consumption relative to exercise intensity

- (i) Define the term 'VO₂ max'.

[2 marks]

- (ii) Explain the change in the shape of the graph **after** the VO₂ max point.

[2 marks]

- (iii) State TWO factors which affect VO₂ max and for EACH give a brief explanation of how it affects the VO₂ max value.

Factor 1:

Factor 2:

[2 marks]

Total 15 marks

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SECTION B

Answer ALL questions.

Write your answers in the spaces provided at the end of each question.

4. Global interest in biodiversity has grown rapidly in recent decades, in parallel with the growing concern about the need for conservation.

(a) (i) Explain the concept of 'biodiversity'. **[2 marks]**

(ii) Discuss why biodiversity loss is a major concern. In your discussion, cite ONE direct and TWO indirect benefits of maintaining biodiversity. **[6 marks]**

(b) (i) Distinguish between *in situ* and *ex situ* conservation of biodiversity, giving an example of EACH type of conservation. **[4 marks]**

(ii) Using your knowledge of *ex situ* conservation methods, briefly comment on THREE limitations of using *ex situ* conservation methods to maintain biodiversity. **[3 marks]**

Total 15 marks

Write the answer to Question 4 here.

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5. (a) (i) Give a brief explanation of the concept of 'homeostasis'. **[2 marks]**
- (ii) A homeostatic system consists of at least three key components: a receptor, a set point and an effector. Discuss the roles of these THREE key components in a homeostatic system. **[6 marks]**
- (b) Collecting ducts play an important role in kidney function and the activity of these ducts is under hormonal control.
- (i) Briefly describe the structure of a collecting duct and comment on its role in kidney function. **[4 marks]**
- (ii) Explain how the antidiuretic hormone, ADH, regulates the function of the collecting ducts. **[3 marks]**

Total 15 marks

Write the answer to Question 5 here.

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- 6.** (a) In addition to the lungs, the heart and circulatory system are also affected by cigarette smoking. Comment on TWO major effects of cigarette smoking on the cardiovascular system of heavy smokers. **[4 marks]**
- (b) (i) Describe the origin and maturation of T lymphocytes. **[3 marks]**
- (ii) State the THREE types of T-cells and for EACH type outline its main function. **[6 marks]**
- (iii) With reference to HIV, explain why T-cells are sometimes unable to effectively eliminate foreign organisms. **[2 marks]**

Total 15 marks

Write the answer to Question 6 here.

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Write the answer to Question 6 here.

[illegible]

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.